

DEPARTMENT OF ECONOMICS

ECONOMICS 114 A2

25 May 2023

Feedback Section C

QUESTION 26

In general, question 26.1 was answered well, while very few students got full marks for question 26.2. In 26.1, Jane's best response is to accept the offer of R300, since if she declined the offer, both Sindisiwe and Jane will get nothing. The answer to question 26.2 is discussed in the textbook in Unit 4 in section 4.10: Dividing a pie (or leaving it on the table). The factors that will influence a Respondent's decision of whether to accept or decline the offer made by the Proposer, are factors such as **social norms**, the Respondent's perception of **fairness**; whether the Respondent cares about **reciprocity** (i.e., to have a preference to be kind and helpful to those who are kind and helpful to them, and to withhold kindness and helpfulness from those that are not kind and helpful to you); **altruism** (caring about others, not only yourself) and **spite** (the decision to decline an offer that is too low to get back at the Proposer).

QUESTION 27

Most students did well in this question, although some students made calculation errors in question 27.2. The correct answer for question 27.1 is **True**. The reason is that the slope of the reservation indifference curve (the MRS) is greater (or steeper) than the slope (or the MRS) of the biological survival constraint. In question 27.2, it is important to note that the total economic rent is maximised where the vertical distance between Ben and Chris's combined feasible frontier is at a maximum. This happens when Ben has **14 hours of free time**, which means that **Ben is working for 10 hours** (24 hours minus the 14 hours of free time). The total economic rent is calculated as the length of the distance between points A and B. At point A, 10 bushels of grain are produced and Ben is consuming 2.5 bushels of grain (at point B). This means that the length between points A and B = $10 - 2.5 = \underline{\underline{7.5 \text{ bushels of grain}}}$.

QUESTION 28

Most students unnecessarily lost marks in this question due to giving incomplete or vague answers. The question guided you by saying that you had to refer to the impact on Ntsako's employment rent in your answer. It is important to note that when Ntsako earns R100 per hour, he spends some of his time reading (instead of working). If CapeNews increases his wage, his employment rent (i.e., **Wage** minus **Reservation wage**) increases. This makes it "more expensive" for Ntsako to lose his job, and therefore his motivation will increase and he will work harder (and spend

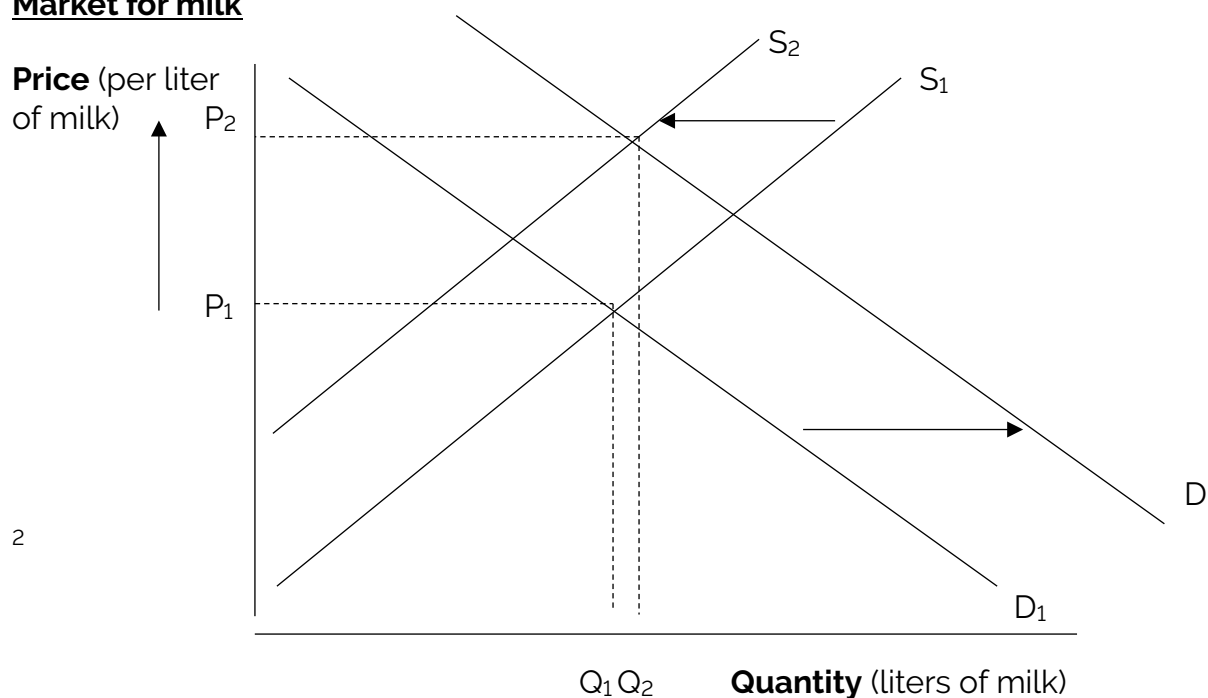
more time working and less time reading). If Ntsako's wage increases (and his employment rent therefore increases) and he is caught out reading at work (instead of working), and if he then loses his job, he is losing a greater amount of employment rent. This motivates him to work harder. In question 28.2, Ntsako's reservation wage will **decrease**. Since the question says that Ntsako loves working with his colleagues on interesting news stories, it means that he is getting a lot of pleasure from his work. This means that he will exert effort from a relatively low level of wage (he doesn't have to be paid much to encourage him to work and exert effort, since he is enjoying his work).

Marks were mostly lost because students did not mention reservation wage in their answer, and only spoke about the worker fearing losing their job, but failed to mention that this is because they value their job more than they used to (i.e. they are no longer as indifferent between keeping and losing their job).

QUESTION 29

There were quite a few mistakes that students made in this question that caused them to lose marks unnecessarily. In quite a few instances, students didn't label the y-axis and x-axis of the graph or didn't label the curves (lines) on the graph. There were also instances that students only drew the graph, but didn't discuss it, which also caused them to lose marks unnecessarily. There were also quite a few students that wrote that the slope of the demand and supply curves change when the curves shift, which is not correct. The slopes of the demand and supply curves do not change when the curves shift. The question asked to evaluate the impact of the two (simultaneous) events on the market for milk.

Market for milk



The market for milk starts off in equilibrium at an equilibrium price of P_1 and an equilibrium quantity of Q_1 . The serious drought (that causes farmers to reduce the

number of cows they keep), means that the farmers will produce less milk. So, the supply of milk will decrease (and the supply curve of milk will shift to the left to S_2). Note that the slope of the supply curve does not change (i.e. the shift is a parallel shift). The medical report that states that milk has health benefits, will increase the demand for milk (people would like to drink more milk to experience these health benefits). Therefore, the demand curve for milk will shift to the right to D_2 (a parallel shift).

It is important to note that we can say that the **equilibrium price of milk will increase**. We are, however, not sure about what the total impact on the equilibrium quantity of milk will be (this depends on how big we draw the shifts in the demand and supply curves). Since the question does not state which shift (supply or demand) is the biggest, we cannot say for certain what will happen to equilibrium quantity. [How the graph is drawn above, shows that the equilibrium quantity of milk will increase, but if we had drawn the supply curve to shift further to the left, then the equilibrium quantity of milk would have decreased. If the size of the shift in the demand and supply curves are of the same size, then the equilibrium quantity of milk will remain unchanged].

Marks were allocated for the graph (3 marks total) and the explanation (2 marks total) that matched what was being indicated in their graph: 1.5 marks total for drawing and explaining supply decreases (shifts left); 1.5 marks total for drawing and explaining demand increases (shifts right); 1 mark for drawing and explaining equilibrium price increases; and 1 mark for drawing and explaining an ambiguous change in quantity (i.e. could have increased, decreased or remained the same). The question indicated that the graph needed to be drawn for the market for milk. Some students misinterpreted the question and drew a graph for a firm in perfect competition, or a monopolist. Marks were also lost where students failed to label their graphs correctly, and where students referred to, for example, an increase in quantity demanded when they were meant to refer to an increase in demand.

QUESTION 30

- 30.1 Fixed cost = R250 (the term in the TC function that does not change as Q changes).
- 30.2 Profit maximization is at $MR = MC$ and MR can be derived from the demand function (slope is 2x as steep as the demand curve's slope).
- 30.3 Economic profit = $TR - TC$ or $(P - AC) \times Q$
- 30.4 There were sometimes strange answers here, while the correct answer refers to the positive profit that attracts entrants to the market.

QUESTION 31

This question was well answered, although some students forgot the formulas. Remember that elasticities are unitless, so do away with the %. Because the demand for coffee is price elastic (1.33), it would make sense for the store to discount it, while the demand for books is inelastic (0.65) and therefore it would not make sense to discount them.

QUESTION 32

- 32.1 The size of the unit tax is determined by calculating the difference between the vertical axis intercepts of the supply curves, i.e., $R42 - R15 = R27$.
- 32.2 The price paid by the buyers after the imposition of the tax are determined by using the slope of the supply curve (identical for both supply curves, $13/26000$), and the vertical intercept of the supply curve after tax ($R42$) - substitute values into the linear function's formula to obtain the price of $R55$.
- 32.3 The deadweight loss is the value of the area of the triangle $(0.5 \times b \times h) = 0.5 \times R27 \times 18000 = R243\ 000$.
- 32.4 Students had to indicate that buyers bear the larger tax burden. The motivation had to indicate that demand was relatively more price inelastic than supply (or that the demand curve is steeper than the supply curve, or that supply is relatively more elastic than demand).
- 32.5 Tax revenue = Tax per unit $\times$ Quantity after tax = $R27 \times 26000 = R702\ 000$.

QUESTION 33

- 33.1 A price ceiling is binding when it is lower than the equilibrium price. You just calculate the break-even price ($R110$) and see that it is higher than the ceiling ($R80$) - therefore binding.
- 33.2 Calculate Q_d and Q_s when the ceiling price of $R80$ applies and see if there is a deficit or surplus - in this case a surplus of $90\ 000$.
- 33.3 The producer surplus will become smaller.
- 33.4 Yes, a subsidy could eliminate the deadweight loss. If government awards a subsidy to producers, the supply curve will shift to the right. This means the market equilibrium price will fall and move towards the price floor of $R80$.

The market equilibrium with higher supply could lead to the Pareto efficient output being produced (no DWL/no shortage).

QUESTION 34

34.1 Most students were able to answer this question. The key was to know that the equilibrium condition was $[P=MC]$. Several students did not provide the equilibrium price and only provided the equilibrium quantity.

34.2 Remember that $MSC = MC + MEC$

And $MSC = \text{Demand}$

It is important to establish in the beginning that MC now becomes MSC and includes the external cost.

34.3 Market efficiency was increased when the price increased by increasing the price and reducing output. Or you could say deadweight loss decreases.